

## ORIGINAL ARTICLE

# High-Efficiency Particulate Air Filters for Preventing Wildfire-related Asthma Complications

## A Cost-Effectiveness Study

Amin Adibi<sup>1,2</sup>, Prabjit Barn<sup>4</sup>, Erin M. Shellington<sup>1,3,5</sup>, Stephanie Harvard<sup>1,5</sup>, Kate M. Johnson<sup>1,2,3,5\*</sup>, and Christopher Carlsten<sup>1,3,5,6\*</sup>

<sup>1</sup>Climate Change Health Effects, Adaptation, and Resilience Research Cluster, <sup>2</sup>Respiratory Evaluation Sciences Program, Collaboration for Outcomes Research and Evaluation, Faculty of Pharmaceutical Sciences, and <sup>3</sup>Division of Respiratory Medicine, Faculty of Medicine, University of British Columbia, Vancouver, British Columbia, Canada; <sup>4</sup>Fraser Health Authority, Surrey, British Columbia, Canada; and <sup>5</sup>Legacy for Airway Health and <sup>6</sup>Air Pollution Exposure Lab, Vancouver Coastal Health Research Institute, Vancouver, British Columbia, Canada

ORCID IDs: 0000-0003-2748-4781 (A.A.); 0000-0001-7406-2448 (K.M.J.).

### Abstract

**Rationale:** Air pollution caused by wildfire smoke is linked to adverse health outcomes, especially for people living with asthma.

**Objectives:** To evaluate whether government rebates for high-efficiency particulate air (HEPA) filters, which reduce concentrations of smoke particles indoors, are cost effective in managing asthma and preventing exacerbations in British Columbia (BC), Canada.

**Methods:** We used a Markov model to analyze health states for asthma control, exacerbation severity, and death over a retrospective time horizon of 5 years (2018–2022). Concentrations of wildfire smoke–derived particulate matter with an aerodynamic diameter  $\leq 2.5 \mu\text{m}$  ( $\text{PM}_{2.5}$ ) from the Canadian Optimized Statistical Smoke Exposure Model and relevant literature informed the model. The base-case analysis assumed continuous use of a HEPA filter. Costs and quality-adjusted life-years (QALYs) resulting from varying rebates were computed for each Health Service Delivery Area (HSDA).

**Measurements and Main Results:** In the base-case analysis, HEPA filter use resulted in increased costs of \$83.34 (SE, \$1.03) and increased QALYs of 0.0011 (SE, 0.0001) per person. The average incremental cost-effectiveness ratio among BC HSDAs was \$74,652/QALY (SE, \$3,517), with incremental cost-effectiveness ratios ranging from \$40,509 to \$89,206 per QALY in HSDAs. Across the province, the intervention was projected to prevent 4,418 exacerbations requiring systemic corticosteroids, 643 emergency department visits, and 425 hospitalizations during the 5-year time horizon. A full rebate was cost effective in 1 of the 16 HSDAs across BC. The probability of cost-effectiveness ranged from 0.1% to 74.8% across HSDAs. A \$100 rebate was cost effective in most HSDAs.

**Conclusions:** The cost-effectiveness of HEPA filters in managing wildfire smoke–related asthma issues in BC varies by region. Government rebates up to two-thirds of the filter cost are generally cost effective, with a full rebate being cost effective only in Kootenay Boundary.

**Keywords:** wildfires; asthma; air filters; particulate matter; climate change

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\*Co-senior authors.

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**Author Contributions:** A.A. and C.C. conceptualized the study. P.B. conducted the literature review. S.H., P.B., and E.M.S. designed and led the patient and stakeholder engagement process. A.A. and K.M.J. developed, coded, and populated the model. A.A. ran the analysis, produced results and visualizations, and developed the web application. A.A., E.M.S., and K.M.J. conducted interviews with knowledge users. C.C. provided clinical input and oversight. K.M.J. provided health economics input and oversight. A.A. drafted the initial manuscript. All authors contributed to critical revision of the manuscript.

Correspondence and requests for reprints should be addressed to Christopher Carlsten, M.D., M.P.H., 166 – 1081 Burrard Street, Vancouver, BC, V6Z 1Y6 Canada. E-mail: carlsten@mail.ubc.ca.

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## At a Glance Commentary

### Scientific Knowledge on the

**Subject:** Air pollution from wildfire smoke poses significant health risks, particularly for individuals living with asthma. High-efficiency particulate air (HEPA) filters are effective in reducing particulate matter from wildfire smoke inside houses.

### What This Study Adds to the

**Field:** We evaluated the cost-effectiveness of government-sponsored HEPA filter rebates in British Columbia, Canada. We found that HEPA filter rebates can be a cost-effective intervention for reducing acute asthma complications in wildfire-prone areas of the province.

The number, size, and intensity of wildfires in Canada have increased, particularly in the western province of British Columbia (BC), with the number of days with uncontrolled wildfire in BC expected to double or triple by 2100 (1).

Wildfire smoke is composed of several pollutants, including particulate matter with an aerodynamic diameter  $\leq 2.5 \mu\text{g}$  ( $\text{PM}_{2.5}$ ).  $\text{PM}_{2.5}$  and other air pollutants have been associated with increased respiratory symptoms, hospitalizations, and other adverse health effects in individuals with asthma (2).

People living with asthma are particularly susceptible to air pollution. Asthma exacerbations (also known as *flare-ups* and *acute severe asthma*) are episodes characterized by progressive worsening of cough, wheezing, shortness of breath, and decrease in lung function (3). Severe exacerbations can be fatal and can occur even in patients with well-controlled asthma (3). Previous studies have shown that  $\text{PM}_{2.5}$  from wildfire events can increase the risk of asthma exacerbations (4).

During wildfire smoke events, indoor  $\text{PM}_{2.5}$  concentrations increase as smoke infiltrates into homes and other buildings. Because people typically spend  $>70\%$  of their time in indoor environments (5), indoor air quality is an important contributor to total air pollution exposure. Consequently, interventions that improve

indoor air quality are important to protecting health, particularly during episodes of poor air quality, such as during wildfire smoke events. Portable high-efficiency particulate air (HEPA) filters can reduce indoor concentrations of  $\text{PM}_{2.5}$  (6). These units work by drawing air across a highly efficient filter that traps particles, including  $\text{PM}_{2.5}$ , and releasing filtered air. HEPA and other filters can be portable or be part of a building's heating, ventilation, and air conditioning system, in which case they are often referred to as in-duct filters.

As the climate emergency worsens and continues to affect air quality (7), there is growing consensus among the public health community that using air filters in indoor settings is an important health-protective intervention, particularly for vulnerable people. The government of Canada currently provides tax benefits for the full cost of an air filter, cleaner, or purifier and up to \$1,000 for the purchase of an air conditioner for patients living with chronic diseases who have prescriptions for these devices (8). In 2021, the Canadian government also announced a 25% tax credit for small businesses to upgrade their ventilation systems and purchase portable HEPA filters (9). In BC, the First Nations Health Authority provides portable air filters to communities affected by wildfires (10). We are also aware of two similar programs in the United States: an air filter distribution program for low-income patients with asthma by the Bay Area Air Quality and Management District (11) and a HEPA filter loaner program by the Forest Stewards Guild in Santa Fe, New Mexico (12). However, we are not aware of any formal analysis evaluating the cost-effectiveness of these programs from a health economics perspective.

In this study, we used a decision-analytic model to evaluate the cost-effectiveness of a government-sponsored portable HEPA filter rebate program for improving asthma control and preventing asthma exacerbations caused by wildfire events in BC. Our analysis can serve as a blueprint for evaluating similar climate change adaptation strategies in BC and elsewhere. Some of the results of this study have been previously reported in the form of two abstracts (13, 14).

## Methods

We have reported the results of this study according to the recommendations and best

practices set forth in the Consolidated Health Economic Evaluation Reporting Standards 2022 statement (15).

On the basis of discussions with policy makers and knowledge users in BC, we chose Health Service Delivery Area (HSDA) as the geographical unit of analysis. Our base-case analysis assumes that the provincial government will offer a 100% rebate for portable HEPA filters to all individuals with diagnoses of asthma in BC. We used a retrospective time horizon of 5 years beginning in 2018 to the end of 2022, which was the most recent 5-year time horizon for which the data were available. This retrospective time horizon was necessary as daily projections of future wildfire  $\text{PM}_{2.5}$  concentrations are not available. We assumed that patients on average spent 69.6% of their time at home (and thus could benefit from the air filter for the proportion of time they were at home), on the basis of the time-use information collected in Statistics Canada's General Social Survey (5). The target population was BC residents with diagnoses of asthma, with a starting age of 42 years (the average age of BC residents) (16).

We projected costs in 2023 Canadian dollars and effects as quality-adjusted life-years (QALYs) for patients with and without portable HEPA filters in their homes, and we report results for each HSDA in BC. We also report the number of averted cases of asthma exacerbations using model-projected exacerbation rates and crude asthma prevalence rates from April 1, 2020, to March 31, 2021, for each HSDA obtained from the BC Centre for Disease Control (17). We calculated incremental cost-effectiveness ratios (ICERs) and net monetary benefit and report cost-effectiveness at a willingness-to-pay (WTP) threshold of \$50,000/QALY.

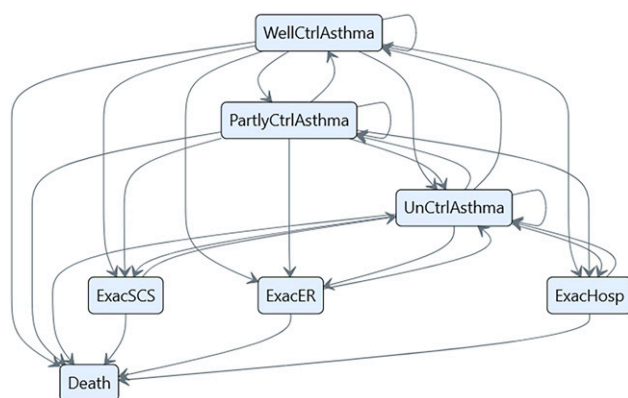
The analysis was conducted from the healthcare payer perspective, with an annual discounting of 1.5% applied to costs and effects.

## Stakeholder Engagement

We developed a health economic analysis plan with early and ongoing input from stakeholders, including two patient partners living with asthma, two medical health officers, an environment health officer, and a policy analyst (*see* ACKNOWLEDGMENT).

## Model Development

We developed a time-varying Markov model with seven health states corresponding to well-controlled asthma, partly controlled



**Figure 1.** Markov health states and transitions. ExacER = exacerbation requiring a visit to the emergency room; ExacHosp = exacerbation requiring hospitalization; ExacSCS = exacerbation requiring systemic corticosteroid use; PartlyCtrlAsthma = partly controlled asthma; UnCtrlAsthma = uncontrolled asthma; WellCtrlAsthma = well-controlled asthma.

asthma, and uncontrolled asthma (as defined per the Global Initiative for Asthma [3]); exacerbations requiring systemic corticosteroid (SCS) use, a visit to the emergency room (ER), or hospitalization; and death (Figure 1).

Background mortality was based on age-specific life tables for BC from Statistics Canada (18). Mortality due to asthma exacerbations of each severity was based on a national review of asthma deaths in the United Kingdom (19, 20). Annual transition probabilities between asthma control states were based on an original analysis of the Economic Burden of Asthma study in which we calculated the proportion of transitions occurring between each control state over five visits conducted over 1 year of follow-up (21). Rates of severe exacerbations leading to

SCS use, ER visit, or hospitalization were obtained from the SYGMA II (A Clinical Study to Evaluate Symbicort Turbuhaler Used “as Needed” in Adults and Adolescents With Asthma) study (22). We applied a risk ratio of 1.40 to individuals with partially controlled and uncontrolled asthma to reflect their higher probability of exacerbation. This parameter was based on an analysis of commercially insured patients in the United States (23).

We ran the model using daily time cycles.

#### Air Pollution Exposure

Average daily outdoor PM<sub>2.5</sub> concentrations were obtained from the Canadian Optimized Statistical Smoke Exposure Model

(CanOSSEM) (24), a random forest machine learning model developed and validated by the BC Centre for Disease Control that projects retrospective average daily wildfire smoke concentrations for each postal code in BC. Outdoor PM<sub>2.5</sub> concentrations in HSDAs were obtained by linking postal codes to HSDAs using Postal Code Conversion File Plus version 7E (25). Model assumptions are listed in Table 1.

Risk ratios for the effect of increased exposure to PM<sub>2.5</sub> on asthma outcomes, including salbutamol dispensation and asthma-related physician visit, ER visit, and hospitalization, were obtained from a recent meta-analysis (4) and a model validation study based on BC administrative health data (26).

Transition probabilities, utility and disutility values, rate ratios for the effect of increased PM<sub>2.5</sub> pollution on asthma outcomes, healthcare state costs, outdoor-to-indoor PM<sub>2.5</sub> infiltration rates, and HEPA filter efficiency rate were obtained from the literature (Table 2).

#### HEPA Filter Effectiveness

We chose what we considered to be a typical HEPA filter unit with a clean-air delivery rate of 105 cfm for smoke and a nominal air exchange rate of 4.8 air changes/h for a coverage area of 15 m<sup>2</sup>. Measured HEPA filter efficiency of 0.31 (defined as the ratio of indoor PM<sub>2.5</sub> measured throughout the year with a HEPA filter to that without a HEPA filter) was obtained from a study led by one

**Table 1.** Model Assumptions for Evaluating the Cost-Effectiveness of a Portable Air Filter Rebate Program to Prevent Asthma Exacerbations

| Assumptions for Base-Case Analysis                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HEPA filters were assumed to operate continuously on their highest setting during the model time horizon.                                                                                                                                                                  |
| We assumed that the government could offer rebates at a discount of 30% of the purchase cost compared with the advertised retail price.                                                                                                                                    |
| We assumed that HEPA filters will need to be replaced every 9 mo of use on the basis of the average manufacturer-recommended timeline, while the filtration unit will need to be replaced every 5 yr regardless of how much it was used.                                   |
| Residents were assumed to cover the costs of electricity and filter replacement.                                                                                                                                                                                           |
| We assumed that people living with asthma received one HEPA filter unit each, even if there were multiple people with asthma in the same home. We assumed that air filters were placed in the main living space or the main bedroom of the person with asthma.             |
| We assumed that people living with asthma spent the same proportion of their day at home as the general population.                                                                                                                                                        |
| Increased salbutamol dispensation (per canister) per 10 µg/m <sup>3</sup> increase in PM <sub>2.5</sub> during wildfire events was used as a proxy for risk of worsened asthma control (i.e., well controlled to partly controlled, or partly controlled to uncontrolled). |
| Potential additional benefits of HEPA filters in reducing exposure to allergens, pathogens, and indoor sources of PM <sub>2.5</sub> such as cooking and wood stoves were not considered.                                                                                   |
| We assumed that all patients would enter the uncontrolled-asthma health state after an exacerbation event.                                                                                                                                                                 |
| Historical wildfire-related PM <sub>2.5</sub> concentrations projected by CanOSSEM were assumed to be accurate.                                                                                                                                                            |

*Definition of abbreviations:* CanOSSEM = Canadian Optimized Statistical Smoke Exposure Model; HEPA = high-efficiency particulate air; PM<sub>2.5</sub> = particulate matter with an aerodynamic diameter ≤2.5 µm.

**Table 2.** Model Parameters

| Parameter                                                                                                  | Base Case                               | DSA              | PSA                | Source                |
|------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------|--------------------|-----------------------|
| Age at start, yr                                                                                           | 42                                      | 33 67            | —                  | (16)                  |
| Mean infiltration efficiency without HEPA filter                                                           | 61%                                     | —                | Normal (SD, 0.27)  | (27)                  |
| Mean infiltration efficiency with HEPA filter                                                              | 19%                                     | —                | Normal (SD, 0.20)  | (27)                  |
| Filter effect                                                                                              | 31%                                     | ±20%             | β                  | (6, 27)               |
| Risk ratio for increased salbutamol dispensation per 10 µg/m <sup>3</sup> increase in PM <sub>2.5</sub>    | 1.04 (1.03–1.06)                        | 1 1.20           | Lognormal          | (26)                  |
| Risk ratio for increased physician visit for asthma per 10 µg/m <sup>3</sup> increase in PM <sub>2.5</sub> | 1.06 (1.04–1.08)                        | 1 1.20           | Lognormal          | (26)                  |
| Risk ratio for asthma-related ER visit per 10 µg/m <sup>3</sup> increase in PM <sub>2.5</sub>              | 1.07 (1.04–1.09)                        | 1 1.20           | Lognormal          | (4)                   |
| Risk ratio for asthma-related hospitalization per 10 µg/m <sup>3</sup> increase in PM <sub>2.5</sub>       | 1.06 (1.02–1.09)                        | 1 1.20           | Lognormal          | (4)                   |
| <b>Probabilities*</b>                                                                                      |                                         |                  |                    |                       |
| Risk of death due to exacerbation (SCS use)                                                                | 0.0267%                                 | —                | β                  | (19, 20)              |
| Risk of death due to exacerbation (ER visit)                                                               | 0.1733%                                 | —                | β                  | (19, 20)              |
| Risk of death due to exacerbation (hospitalization)                                                        | 0.1801%                                 | —                | β                  | (19, 44)              |
| Well-controlled to uncontrolled asthma, monthly                                                            | 1.30%                                   | —                | β                  | (21)                  |
| Well-controlled to partly controlled asthma, monthly                                                       | 13.03%                                  | —                | β                  | (21)                  |
| Partly controlled to well-controlled asthma, monthly                                                       | 10.07%                                  | —                | β                  | (21)                  |
| Partly controlled to uncontrolled asthma, monthly                                                          | 9.04%                                   | —                | β                  | (21)                  |
| Uncontrolled to partly controlled asthma, monthly                                                          | 12.27%                                  | —                | β                  | (21)                  |
| Uncontrolled to well-controlled asthma, monthly                                                            | 3.95%                                   | —                | β                  | (21)                  |
| Annual rate of exacerbation (SCS use) in controlled asthma                                                 | 0.0895<br><i>P</i> = 8.55% <sup>†</sup> | —                | β                  | (22, 23)              |
| Annual rate of exacerbation (ER visit) in controlled asthma                                                | 0.0111<br><i>P</i> = 1.11% <sup>‡</sup> | —                | β                  | (22, 23)              |
| Annual rate of exacerbation (hospitalization) in controlled asthma                                         | 0.0086<br><i>P</i> = 0.85% <sup>§</sup> | —                | β                  | (22, 23)              |
| Risk ratio for exacerbations (SCS use, ER, or hospitalization) in uncontrolled vs. well-controlled asthma  | 1.1127 <sup>  </sup>                    | ±20%             | Lognormal          | (23)                  |
| Risk ratio for exacerbations in partly controlled vs. well-controlled asthma                               | 1.0352 <sup>  </sup>                    | ±20%             | Lognormal          | (23)                  |
| Exacerbation (SCS use, ER, or hospitalization) to uncontrolled asthma                                      | 1 – <i>P</i> <sub>mortality</sub>       | —                | Fixed              | —                     |
| <b>Exposures</b>                                                                                           |                                         |                  |                    |                       |
| Monthly PM <sub>2.5</sub> concentrations (as average for each postal code)                                 | CanOSSEM                                | —                | Fixed              | (24)                  |
| <b>Unit costs</b>                                                                                          |                                         |                  |                    |                       |
| HEPA filter unit                                                                                           | \$150                                   | ±20%             | γ                  | Retail price          |
| Government discount on retail price                                                                        | 30%                                     | —                | Assumption         | —                     |
| Indoor HEPA filter electricity use, annually (when used continuously)                                      | \$9.90                                  | Fixed            | Fixed              | BC Hydro calculator** |
| Filter replacement, per replacement                                                                        | \$30                                    | ±20%             | γ                  | Retail price          |
| Direct costs of well-controlled asthma, monthly                                                            | \$323.57                                | ±20%             | Normal (SD, 59.50) | (45)                  |
| Direct costs of partly controlled asthma, monthly                                                          | \$404.46                                | ±20%             | Normal (SD, 27.41) | (45)                  |
| Direct costs of uncontrolled asthma, monthly                                                               | \$426.56                                | ±20%             | Normal (SD, 34.82) | (45)                  |
| Exacerbation (SCS use)                                                                                     | \$181.71                                | \$138 \$208      | γ                  | (28)                  |
| Exacerbation (ER visit)                                                                                    | \$574.88                                | \$438 \$657      | γ                  | (28)                  |
| Exacerbation (hospitalization stay unit)                                                                   | \$11,009.89                             | \$8,389 \$12,583 | γ                  | (28)                  |
| <b>Utilities</b>                                                                                           |                                         |                  |                    |                       |
| Utility of controlled asthma, daily                                                                        | 0.70/365                                | ±20%             | β                  | (45)                  |
| Utility of partly controlled asthma, daily                                                                 | 0.66/365                                | ±20%             | β                  | (45)                  |
| Utility of uncontrolled asthma, daily                                                                      | 0.61/365                                | ±20%             | β                  | (45)                  |
| Disutility of exacerbations (SCS use), per event                                                           | 0.0057                                  | –0.08 –0.12      | Normal (SD, 0.01)  | (29)                  |
| Disutility of exacerbations (ER visit), per event                                                          | 0.00745                                 | –0.12 –0.18      | Normal (SD, 0.015) | Assumption            |
| Disutility of exacerbations (hospitalization), per event                                                   | 0.0092                                  | –0.16 –0.24      | Normal (SD, 0.02)  | (29)                  |
| <b>Other parameters</b>                                                                                    |                                         |                  |                    |                       |
| Proportion of time spent at home                                                                           | 69.6%                                   | —                | Fixed              | (5)                   |
| Discounting (annual)                                                                                       | 1.5%                                    | 0% 5%            | —                  | —                     |

(Continued)

Table 2. (Continued)

| Parameter                     | Base Case | DSA | PSA   | Source     |
|-------------------------------|-----------|-----|-------|------------|
| Air filter unit life span, yr | 5         | —   | Fixed | —          |
| HEPA filter life span, mo     | 9         | —   | Fixed | Assumption |

*Definition of abbreviations:* CanOSSEM = Canadian Optimized Statistical Smoke Exposure Model; DSA = deterministic sensitivity analysis; ER = emergency room; HEPA = high-efficiency particulate air; PM<sub>2.5</sub> = particulate matter with an aerodynamic diameter ≤2.5 μm; PSA = probabilistic sensitivity analysis; SCS = systemic corticosteroid.

\*Annual and monthly probabilities were rescaled to daily probabilities using  $p_m = 1 - (1 - p)^{(1/n)}$ .

†Per Table 1 in Bateman and colleagues (22), 45.9% of patients had uncontrolled asthma, and 54.1% had controlled asthma. Per Bateman and colleagues, the overall annual rate of exacerbations requiring SCS use was 209 of 1,998. We combined this information with risk ratios for degree of asthma control and exacerbations from Pollack and colleagues (23), who reported mean annual exacerbation rates of 1.60, 1.75, and 2.19 for patients with well-controlled, partly controlled, and uncontrolled asthma, respectively, to solve for the annual exacerbation risks for those with well controlled asthma:  $0.541 \times R_{\text{exacSCS-wellCtrl}} + 0.459 \times R_{\text{exacSCS-wellCtrl}} \times 1.36875 = \frac{209}{1998}$  so  $R_{\text{exacSCS-wellCtrl}} \approx 0.0894625$ .

‡Per Table 1 in Bateman and colleagues (22), 45.9% of patients had uncontrolled asthma, and 54.1% had controlled asthma. Per Bateman and colleagues, the overall annual rate of exacerbations requiring an ER visit was 26 of 1,998. We combined this information with risk ratios for degree of asthma control and exacerbations from Pollack and colleagues (23), who reported mean annual exacerbation rates of 1.60, 1.75, and 2.19 for patients with well-controlled, partly controlled, and uncontrolled asthma, respectively, to solve for the annual exacerbation risks for those with controlled asthma:  $0.541 \times R_{\text{exacER-wellCtrl}} + 0.459 \times R_{\text{exacER-wellCtrl}} \times 1.36875 = \frac{26}{1998}$  so  $R_{\text{exacER-wellCtrl}} \approx 0.0111293$ .

§Per Table 1 in Bateman and colleagues (22), 45.9% of patients had uncontrolled asthma, and 54.1% had controlled asthma. Per Bateman and colleagues, the overall annual rate of exacerbations requiring hospitalization was 20 of 1,998. We combined this information with risk ratios for degree of asthma control and exacerbations from Pollack and colleagues (23), who reported mean annual exacerbation rates of 1.60, 1.75, and 2.19 for patients with well-controlled, partly controlled, and uncontrolled asthma, respectively, to solve for the annual exacerbation risks for those with controlled asthma:  $0.541 \times R_{\text{exacHosp-wellCtrl}} + 0.459 \times R_{\text{exacHosp-wellCtrl}} \times 1.36875 = \frac{20}{1998}$  so  $R_{\text{exacHosp-wellCtrl}} \approx 0.00856101$ .

||Calculated from reported rates for exacerbation. Rates of 2.19 and 1.6 for exacerbations in patients with poorly controlled and well-controlled asthma were converted to risk probabilities of 0.8881 and 0.7981, respectively.

¶Calculated from reported rates for exacerbation. Rates of 1.75 and 1.6 for exacerbations in patients with partly controlled and well-controlled asthma were converted to risk probabilities of 0.8262 and 0.7981, respectively.

\*\*Calculations are based on a Blue Pure 411 Auto (Blueair, Stockholm, Sweden) unit running at the highest setting (10W) for 24 hours every day for a year (87.60kWh) at an average residential rate of 11.30 cents/kWh (a blend of step 1 and step 2 rates). For more information, refer to <https://www.bchydro.com/powersmart/residential/tools-and-calculators/cost-calculator.html>.

of our coauthors that evaluated air filter effectiveness in BC homes during smoke events (27) using a comparable air filter unit with a clean-air delivery rate of 150 cfm and a nominal air exchange rate of 6 air changes/h for a coverage area of 17.37 m<sup>2</sup> (Table 2). Varying filter effectiveness values of ±20% were explored in one-way sensitivity analysis.

Costs

Costs included the initial purchase price of the HEPA filter unit, background healthcare costs based on the degree of asthma control, and unit costs of exacerbations obtained from the literature (22, 28).

Costs to patients for air filter operation such as electricity and replacement HEPA filters after every 9 months of use (based on average replacement duration according to the manufacturer) were not included in the cost-effectiveness analysis but are reported in the online supplement.

Health-State Utilities

Health-state utilities were derived from the literature on the basis of degrees of asthma control, while severe exacerbations requiring

SCS use, ER visit, or hospitalization were associated with a one-time disutility value derived from EQ-5D questionnaires (29).

Sensitivity Analyses

One-way deterministic sensitivity analysis was used to explore the effect of changing assumptions on the estimated costs and QALYs. Uncertainty in the results due to parameter uncertainty was explored through probabilistic sensitivity analysis with 1,000 iterations from parameter distributions (Table 2) in each HSDA.

Our base-case scenario assumed that the government covered the full cost of the air filter and that air filters were operating continuously throughout the five years of study. Additionally, we explored three different scenarios: 1) the government pays a \$100 (67%) rebate; 2) the government pays a full (100%) rebate, and air filters are turned on only when the outdoor pollution exceeds certain thresholds; and 3) the government pays a \$30 (20%) rebate, and the air filter operates only when outdoor PM<sub>2.5</sub> concentration is above a certain threshold. We chose rebate amounts on the basis of convenience and existing provincial rebate

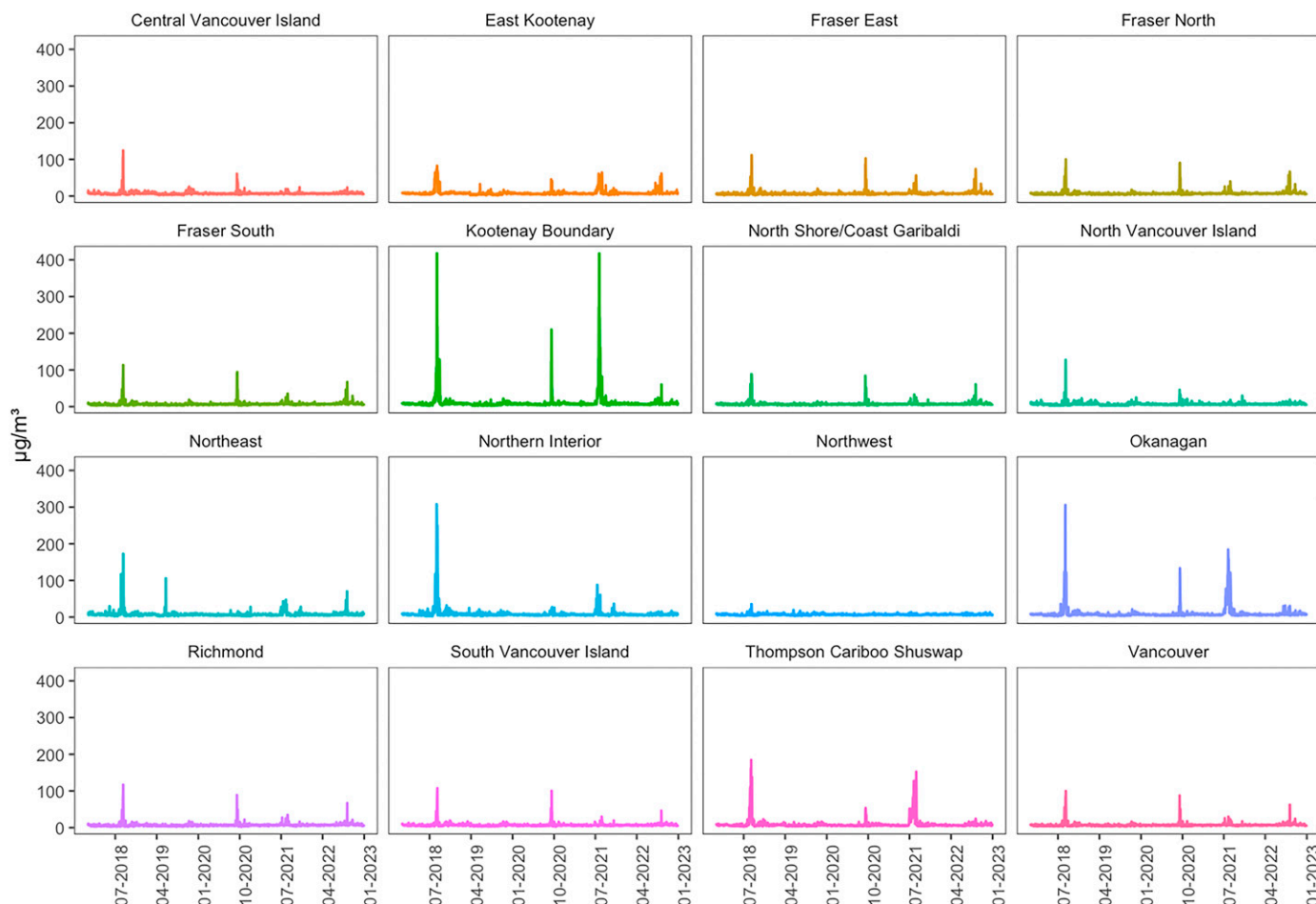
programs (e.g., for energy-efficient products (30)).

Software

Data preparation, model development, and statistical analysis was performed in R version 4.3.1 (<https://www.r-project.org>) using the heemod package version 0.15.1 (31). We used Quarto version 1.4.346 (<https://quarto.org>) to create a reproducible manuscript and used version control to keep track of methodological decisions and changes to the model. Model code is publicly available at [https://github.com/resplab/hepa\\_wildfire\\_CE\\_code](https://github.com/resplab/hepa_wildfire_CE_code).

Results

Average daily wildfire-related smoke concentration ranged from 2.5 μg/m<sup>3</sup> (September 25, 2019, Northeast) to 410.6 μg/m<sup>3</sup> (August 19, 2018, Kootenay Boundary). Significant year-to-year variability was observed among all HSDAs, with higher smoke concentration during years with more wildfire activity in the Interior and Northern Health regions, as shown in Figure 2.



**Figure 2.** Daily smoke exposure (particulate matter with an aerodynamic diameter  $\leq 2.5 \mu\text{m}$ ) across British Columbia.

### Base-Case Cost-Effectiveness

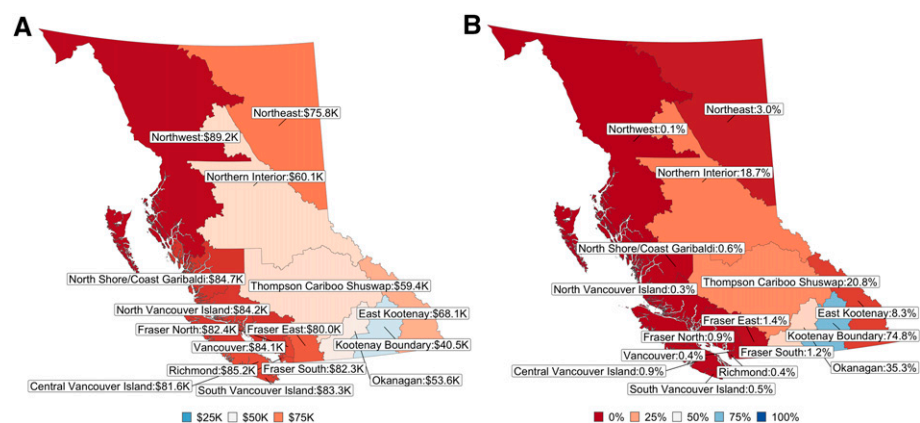
Figure 3 shows the ICER for each HSDA in BC during the time horizon and the associated probability of cost-effectiveness

when the uncertainty around model input parameters (Table 2) is taken into account. In the base-case analysis in which the government paid 100% of the purchase cost

for HEPA filter units, the ICER was below a WTP threshold of \$50,000/QALY in Kootenay Boundary and above the threshold elsewhere in the province.

Table 3 ranks HSDAs in BC in terms of HEPA rebate program cost-effectiveness in descending order on the basis of ICER. ICERs ranged from \$40,509/QALY in Kootenay Boundary to \$89,206/QALY in Northwest. On the basis of model projections and the prevalence of asthma in BC, a total of 4,418 severe exacerbations leading to SCS use, 643 ER visits, and 425 hospitalizations could be averted by continuous HEPA filter use. Because of the larger populations and higher prevalence of asthma, the greatest number of severe exacerbations averted (including SCS use, ER visits, and hospitalizations) were in Fraser South (961), Fraser North (644), Okanagan (607), and Vancouver (590).

Cost-effectiveness probabilities were highest in the Kootenay Boundary (74.8%),



**Figure 3.** Base-case results. (A) Incremental cost-effectiveness ratios for the 100% high-efficiency particulate air filter rebate program. (B) Cost-effectiveness probability at a willingness to pay of \$50,000/quality-adjusted life-year.

**Table 3.** Incremental Cost-Effectiveness Ratios for the Portable High-Efficiency Particulate Air Cleaner Rebate Program in British Columbia

| HSDA                        | $\Delta$ Cost | $\Delta$ QALYs | ICER     | $\Delta$ Exacerbation |     |      | $P_{CE}$ | NMB     |
|-----------------------------|---------------|----------------|----------|-----------------------|-----|------|----------|---------|
|                             |               |                |          | SCS Use               | ER  | Hosp |          |         |
| Kootenay Boundary           | \$71.6        | 0.0018         | \$40,509 | 112                   | 17  | 11   | 74.8%    | \$16.9  |
| Okanagan                    | \$77.8        | 0.0014         | \$53,621 | 488                   | 72  | 47   | 35.3%    | -\$5.2  |
| Thompson Cariboo Shuswap    | \$79.8        | 0.0013         | \$59,428 | 290                   | 42  | 28   | 20.8%    | -\$12.8 |
| Northern Interior           | \$80.1        | 0.0013         | \$60,119 | 165                   | 24  | 16   | 18.7%    | -\$13.6 |
| East Kootenay               | \$82.3        | 0.0012         | \$68,064 | 68                    | 10  | 6    | 8.3%     | -\$21.8 |
| Northeast                   | \$84.2        | 0.0011         | \$75,804 | 52                    | 8   | 5    | 3.0%     | -\$28.7 |
| Fraser East                 | \$85.0        | 0.0011         | \$79,975 | 348                   | 51  | 33   | 1.4%     | -\$32.0 |
| Central Vancouver Island    | \$85.4        | 0.0010         | \$81,566 | 268                   | 39  | 26   | 0.9%     | -\$32.9 |
| Fraser South                | \$85.5        | 0.0010         | \$82,259 | 775                   | 112 | 74   | 1.2%     | -\$33.5 |
| Fraser North                | \$85.5        | 0.0010         | \$82,377 | 519                   | 75  | 50   | 0.9%     | -\$33.5 |
| South Vancouver Island      | \$85.7        | 0.0010         | \$83,319 | 335                   | 49  | 32   | 0.5%     | -\$34.2 |
| Vancouver                   | \$85.8        | 0.0010         | \$84,127 | 475                   | 69  | 46   | 0.4%     | -\$34.9 |
| North Vancouver Island      | \$85.9        | 0.0010         | \$84,164 | 116                   | 17  | 11   | 0.3%     | -\$34.9 |
| North Shore/Coast Garibaldi | \$86.0        | 0.0010         | \$84,691 | 217                   | 31  | 21   | 0.6%     | -\$35.5 |
| Richmond                    | \$86.0        | 0.0010         | \$85,201 | 132                   | 19  | 13   | 0.4%     | -\$35.5 |
| Northwest                   | \$86.8        | 0.0010         | \$89,206 | 58                    | 8   | 6    | 0.1%     | -\$38.3 |

*Definition of abbreviations:* ER = emergency room; Hosp = hospitalization; HSDA = Health Service Delivery Area; ICER = incremental cost-effectiveness ratio; NMB = net monetary benefit;  $P_{CE}$  = probability of cost-effectiveness on the basis of probabilistic sensitivity analysis; QALY = quality-adjusted life-year; SCS = systemic corticosteroid.

Okanagan (35.3%), and Thomson Cariboo Shuswap (20.8%) HSDAs. One-way sensitivity analysis (see the online supplement) showed that costs and QALYs were most sensitive to the risk ratios of increased salbutamol dispensation and hospitalization per 10  $\mu\text{g}/\text{m}^3$  increase in  $\text{PM}_{2.5}$ , the utility of well-controlled and uncontrolled asthma, and the retail price of air filter units.

### Scenario Analyses

Figure 4 shows the results of scenario analyses. Our results suggest that a \$100 rebate program would have been cost effective at a WTP threshold of

\$50,000/QALY everywhere in the province except for the North Shore/Coast Garibaldi and Northwest HSDAs, with ICERs of \$50,500/QALY and \$53,200/QALY, respectively.

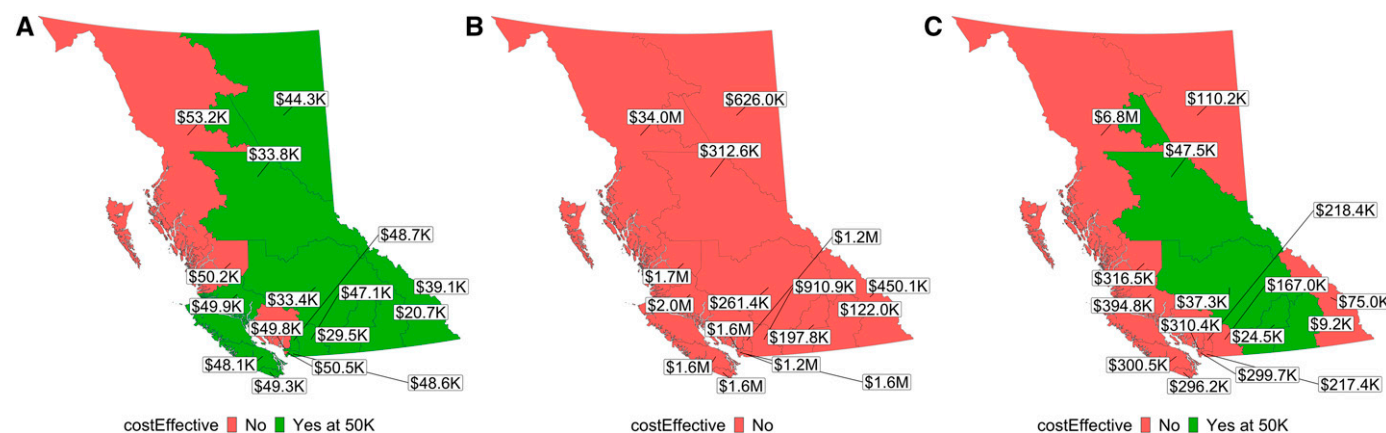
The next two scenarios are based on the operation of HEPA filters when outdoor  $\text{PM}_{2.5}$  exceeded a threshold concentration. We used a threshold of 25  $\mu\text{g}/\text{m}^3$  for  $\text{PM}_{2.5}$ , on the basis of the BC government's 24-hour ambient air quality objective, which is used, together with other information, to guide decisions on when to issue an air quality advisory (32).

Days with  $\text{PM}_{2.5}$  concentrations above 25  $\mu\text{g}/\text{m}^3$  were most common in August,

followed by September, July, October, and May. Our results suggest that a full purchase rebate together with operation of air filters on days when outdoor  $\text{PM}_{2.5}$  concentrations exceeded 25  $\mu\text{g}/\text{m}^3$  would not have been cost effective anywhere in BC.

The last scenario considered a combination of a \$30 rebate and operation of air filters on days when  $\text{PM}_{2.5}$  concentrations exceeded 25  $\mu\text{g}/\text{m}^3$ . Our results suggest that the intervention would have been cost effective in Kootenay Boundary, Okanagan, Thompson Cariboo Shuswap, and Northern Interior.

Other possible scenarios and the effect of alternative inputs on the results can be



**Figure 4.** Incremental cost-effectiveness ratios for the high-efficiency particulate air filter rebate program: different scenarios. (A) Scenario 1: \$100 rebate. (B) Scenario 2: full rebate and air filters on when  $\text{PM}_{2.5} \geq 25 \mu\text{g}/\text{m}^3$ . (C) Scenario 3: \$30 rebate and air filters on when  $\text{PM}_{2.5} \geq 25 \mu\text{g}/\text{m}^3$ .  $\text{PM}_{2.5}$  = particulate matter with an aerodynamic diameter  $\leq 2.5 \mu\text{m}$ .

explored further using a web application, available at [https://resplab.shinyapps.io/hepa\\_wildfire\\_CE/](https://resplab.shinyapps.io/hepa_wildfire_CE/).

### Operation Costs

Although a formal evaluation of the intervention from a societal perspective is beyond the scope of this work, operation costs for patients were calculated to provide additional context. In the base-case analysis when the air filter is operating continuously at its highest setting, patients anywhere in BC can expect to pay an average of \$10 for 87.60 kWh of electricity and \$40 for HEPA filter replacements annually, for a total of \$50 per year. In the threshold-based scenarios, operation costs would be much lower (between \$0.03 and \$1.91) and across different HSDAs, as shown in Table E1.

### Discussion

We found that across BC, offering a 100% rebate on HEPA filters was cost effective between 2018 and 2022 in the Kootenay Boundary HSDA, which was the most wildfire-prone HSDA in that time frame. Our results suggest that a \$100 rebate program was cost effective in most of the province when air filters were used continuously throughout the year. When air filters are operated only on days when  $PM_{2.5}$  concentrations exceed  $25 \mu g/m^3$ , a \$30 rebate program was also cost effective in wildfire-prone areas of the interior and northern interior of BC. To the best of our knowledge, this is the first cost-effectiveness analysis of a government-sponsored HEPA filter rebate program designed to prevent wildfire smoke-related asthma exacerbations and improve asthma control.

Particulate matter pollution is a major cause of health and economic burden in Canada. In its 2022 report on the health of Canadians in a changing climate, Health Canada classified fine particulate matter among the three major outdoor pollutants that are collectively responsible for 15,300 premature deaths in Canada annually, with an economic cost of \$114 billion (33).

There are growing calls for governments to better protect health, including by covering the cost of climate adaptation measures that protect the public. For example, the BC Coroners Service report on the 2021 heat dome in BC, which resulted in 619 deaths, recommended that the BC government increase the accessibility of air conditioners

for use during extreme events by allowing them to be provided as medical devices through existing provincial programs (34). In response to this report, the BC government launched a new initiative in June 2023 to provide 8,000 publicly funded air conditioning units to low-income and medically vulnerable individuals (35). Heat events and smoke can occur together, and the current public health advice is to create or access cool environments with clean air. Our results suggest that a similar program should be implemented for HEPA filter air cleaners to mitigate the impacts of extreme wildfire events in HSDAs with recurrently high wildfire smoke exposure. Considering the equity implications of such programs, we believe that offering rebates for portable HEPA filters can enhance equal access to healthier indoor environments. Such rebates could extend affordability to renters too, as presently available rebates primarily target homeowners.

We made several assumptions to develop our cost-effectiveness model. Where possible, we opted for assumptions that would minimize the chance of wrongly identifying the intervention as cost effective. For instance, we narrowly focused on the short-term health benefits of HEPA filters in preventing acute asthma complications. However, chronic exposure to wildfire smoke may also be associated with increased risk of asthma incidence. Maintaining asthma control and preventing exacerbations is likely associated with improved long-term respiratory outcomes, which were not accounted for in our analysis. We considered the benefits of air filters only in reducing exposure to wildfire-related  $PM_{2.5}$ . However, HEPA filters reduce concentrations of  $PM_{2.5}$  from all sources, including traffic and industry, indoor sources, allergens, bacteria, and respiratory viruses such as flu and coronavirus disease 2019 (COVID-19). We also assumed that HEPA filter units would last only 5 years, regardless of how much they were in use, while the HEPA filters had to be replaced every 9 months.

We assumed that individuals with asthma spent the same proportion of time indoors as the general public. However, it is plausible that people living with asthma might increase their time indoors on days with high concentrations of wildfire pollution, thereby improving the cost-effectiveness of portable HEPA filters compared with what we have reported.

In our base-case analysis, we assumed the air filter to be turned on continuously for

the 5-year time horizon of the model, which is in line with Health Canada's guideline asserting that there is no threshold of exposure to  $PM_{2.5}$  at which negative health effects may not occur (36). Continuous operation of air filters also ensures further benefits from reducing exposure to indoor sources of  $PM_{2.5}$  and allergens and reduces transmission of respiratory infections.

There might be concerns about the practicality of running portable HEPA filters continuously. Previous studies have shown that adherence might be negatively affected because of the machine's noise and the perceived cold draft from the machines, especially during winter (37). Our study implicitly accounts for this, as we have relied on real-world experimental measurements of filter effect that were done in summer and winter across BC (27).

CanOSSEM provides estimates for  $PM_{2.5}$  in general, with improving accounting for wildfire smoke. Therefore, our results reflect the impact of HEPA filters on  $PM_{2.5}$  attributable to all sources, although in the Pacific Northwest, wildfires are the biggest contributor to  $PM_{2.5}$  (38–40). Our scenario analyses also showed that continuous operation of the HEPA filter is more beneficial than turning it on and off daily on the basis of the provincial 24-hour  $PM_{2.5}$  ambient air quality objective. It makes sense for continuous operation to be the most cost-effective choice from the government's perspective, as there is more benefit to reap with no additional cost because the government is paying only for the upfront cost of a rebate.

Several limitations should be noted. First, the stochastic and hard-to-predict nature of wildfire events prevented us from conducting this analysis prospectively, as the long-term prediction of wildfire events in BC with adequate spatial and temporal resolution is not available. Our retrospective results are still useful for future planning, as the frequency and intensity of wildfires in BC is expected to grow, and higher degrees of exposure will make the intervention more cost effective.

Second, retrospective wildfire-related  $PM_{2.5}$  concentrations used in this study are based on the results of CanOSSEM, and thus subject to limitations and uncertainties of that model.

Third, within the observed  $PM_{2.5}$  concentration range of  $2.3$ – $417.3 \mu g/m^3$ , we have assumed a linear dose-response relationship for increased risk of change in

asthma control and asthma exacerbations leading to SCS use, ER visit, or hospitalization.

Last, because of a lack of data, we did not evaluate HEPA filters in subgroups of the population on the basis of sex, age, ethnicity, or social determinants of health, despite their established impact on the burden of the disease (41–43).

## Conclusions

Between 2018 and 2022, offering a 100% rebate on portable HEPA filters was a cost-effective intervention to reduce short-term asthma complications due to wildfire

smoke in Kootenay Boundary but not in other HSDAs in BC. Consumer rebates of up to \$100 (about two-thirds of the cost of the air filter unit) were a cost-effective alternative in most of the province, especially the interior and northern interior parts of the province, where wildfire exposure is higher. ■

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